*Astronomy & Astrophysics* manuscript no. aanda
April 16, 2025 

# A possible wave-optical effect in lensed FRBs

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April 16, 2025

**ABSTRACT**

*Context.* Fast Radio Bursts (FRBs) are enigmatic extragalactic bursts whose properties are still largely unknown, but based on their extremely small time duration, they are proposed to have a compact structure, making them candidates for wave-optical effects if gravitational lensed. If an FRB is lensed into multiple-images bursts at different times by a galaxy or cluster, a likely scenario is that only one image is detected, because the others fall outside the survey area and time frame.

*Aims.* In this work we explore the FRB analog of quasar microlensing, namely the collective microlensing by stars in the lensing galaxy, now with wave optics included. The eikonal regime is applicable here.

*Methods.* We study the voltage (rather than the intensity) in a simple simulation consisting of (a) microlensing stars, and (b) plasma scattering by a turbulent interstellar medium.

*Results.* The auto-correlation of the voltage shows peaks (at order-microsecond separations) corresponding to wave-optical interference between lensed micro-images. The peaks are frequency dependent if plasma-scattering is significant. While qualitative and still in need of more realistic simulations, the results suggest that a strongly-lensed FRB could be identified from a single image.

*Conclusions.* Microlensing could sniff out macro-lensed FRBs.

**Key words.** fast radio bursts – strong lensing



## 1. Introduction

Ever since the discovery of an unprecedented signal in an archival radio time series by D. J. Narkevic, then an undergraduate researcher, and the interpretation of that signal as an exceptionally bright extragalactic radio transient (Lorimer et al. 2007), Fast Radio Bursts (FRBs) have steadily increased in interest (Ng 2023; Lorimer et al. 2024).

FRBs last from microseconds to milliseconds, and are characterized by very high dispersion measures indicating cosmological distances, which implies that they are 10 to 12 orders of magnitude brighter than radio sources within the Milky Way. Also, they are very common across the sky, with a FRB expected once every ten seconds (Keane & Petroff 2015; Champion et al. 2016). While early single-dish telescopes like Parkes and Arecibo could detect only a few FRBs, the CHIME telescope (CHIME/FRB Collaboration et al. 2018) with its much larger field of view has detected hundreds (CHIME/FRB Collaboration et al. 2021). Observation of additional signals from FRB121102 led to the understanding that FRBs can be of a repeating nature (Spitler et al. 2014, 2016), showing that they are not always cataclysmic events. While follow-up observations with single-dish telescopes were difficult, CHIME has reported that around 10% of FRBs detected have a repeating feature (Andersen et al. 2020; Andersen et al. 2023), and a subset of those have periodic activity windows. Repeating FRBs have wide-ranging properties, peaking at different frequencies, with the presence of multiple components (e.g., Chatterjee et al. 2017; Tendulkar et al. 2017). In many cases, precise localization can be carried out, and in these cases, the studies of the intergalactic medium are possible.

The rapid time variation in FRBs indicates small intrinsic sizes. This would imply that FRB signals are spatially coherent, which would lead to interference effects in cases of gravitational lensing. Several authors (Eichler 2017; Katz et al. 2020; Kader et al. 2022; Leung et al. 2022; Kumar & Beniamini 2023) have considered the possible lensing of FRBs by black holes or other compact masses. These works also note a significant complication, which is frequency dependent plasma lensing by the interstellar medium in the host galaxy and in the Milky Way.

Wucknitz et al. (2021) consider multiple images in galaxy and cluster lensing, and draw attention to the prospect of measuring redshift drift if time delays from FRB repeaters are observed. One of the potential complications noted in that work is the collective microlensing by stars in the lensing galaxy or cluster. Lewis (2020) studied the expected effect of microlensing on lensing time delays. A difficulty in finding such multiple-image systems is that a survey may detect one image from a system, but miss the others because the survey pointing has moved to a different part of the sky. Since of order one in a thousand high-redshift objects is lensed into multiple images, it is plausible that a strongly-lensed FRB has already been observed without being recognisable as such.

In this work we carry out some simple simulations of wave optics in collective microlensing by stars. That is, we combine ideas from the works mentioned in the previous two paragraphs. The result suggests that collective microlensing by stars imprints